Training accuracy = 99%
Test " = 75%

Bias → too simple

Variance → too sensitive

High bias → Underfitting
High variance → Overfitting

Bias	Variance	What happens?
High	Low	Underfitting
Low	High	Overfitting
High	High	Bad
Low	Low	Ideal model

Simple model → more complex → more complex → very complex
Bias decrease
variance increase

Bias - variance trade off.

Decision Tree

tree depth = 1

Income
Yes / No ?

High Bias + Low variance.
→ Underfitting.

tree depth = 5

Income
Age / Expense

Lower Bias + Reasonable variance.
→ Good generalization.

Regularization :-

Model A	Model B
$w_1 = 2$	$w_1 = 100$
$w_2 = 3$	$w_2 = -200$
$w_3 = 4$	$w_3 = 150$

Normal training
Loss = prediction error.

Regularized training
Loss = predi. error + penalty

Ridge / L_2
Large coefficients → Shrink them

Lasso / L_1
Large coefficients. → Shrink them

Ridge

$$MSE + \lambda \sum_{j=1}^{p} \beta_j^2$$

Prediction error
regularization strength

$\boxed{\lambda = 0}$

$\boxed{\text{error} = MSE}$

↑ λ

$100x_1 + 80x_2 + 70x_3 + 5$

$40x_1 + 30x_2 + 20x_3 + 5$

0%

Lasso

$$MSE + \lambda \sum_{j=1}^{p} |\beta_j|$$

$\beta_j = 0$

Elastic Net = $L_1 + L_2$

$$MSE + \lambda_1 \sum |\beta_j| + \lambda_2 \sum |\beta_j|^2$$